

Transactions of the American Society of Civil Engineers, Vol. LXX, Dec. 1910

B. S. Randolph

Project Gutenberg

Transactions of the American Society of Civil Engineers, Vol. LXX, Dec. 1910

B. S. Randolph

Project Gutenberg

The Project Gutenberg eBook of Transactions of the American Society of Civil Engineers, Vol. LXX, Dec. 1910

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Transactions of the American Society of Civil Engineers, Vol. LXX, Dec. 1910

Author: B. S. Randolph

Release date: July 3, 2006 [eBook #18747]

Language: English

Other information and formats: www.gutenberg.org/ebooks/18747

Credits: Produced by Juliet Sutherland, Sigal Alon and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK TRANSACTIONS OF THE AMERICAN SOCIETY OF
CIVIL ENGINEERS, VOL. LXX, DEC. 1910 ***

AMERICAN SOCIETY OF CIVIL ENGINEERS

INSTITUTED 1852

TRANSACTIONS

Paper No. 1172

**LOCOMOTIVE PERFORMANCE ON GRADES OF
VARIOUS LENGTHS.**

BY BEVERLY S. RANDOLPH, M. AM. SOC. C. E.

**WITH DISCUSSION BY MESSRS. C. D. PURDON, JOHN C. TRAUTWINE, JR., AND
BEVERLY S. RANDOLPH.**

In the location of new railways and the improvement of lines already in operation, it is now well recognized that large economies can be effected by the careful study of train resistance due to grades and alignment, distributing this resistance so as to secure a minimum cost of operation with the means available for construction.

While engaged in such studies some years ago, the attention of the writer was attracted by the fact that the usual method of calculating the traction of a locomotive—by assuming from 20 to 25% of the weight on the drivers—was subject to no small modification in practice.

In order to obtain a working basis, for use in relation to this feature, he undertook the collection of data from the practical operation of various roads. Subsequent engagements in an entirely different direction caused this to be laid aside until the present time. The results are given in [Table 1](#), from which it will be seen that the percentage of driver weight utilized in draft is a function of the length as well as the rate of grade encountered in the practical operation of railways.

In this table, performance will be found expressed as the percentage of the weight on the drivers which is utilized in draft. This is calculated on a basis of 6 lb. per ton of train resistance, for dates prior to 1880, this being the amount given by the late A. M. Wellington, M. Am. Soc. C. E.,^[A] and 4.7 lb. per ton for those of 1908-10, as obtained by A. C. Dennis, M. Am. Soc. C. E.,^[B] assuming this difference to represent the advance in practice from 1880 to the present time. Most of the data have been obtained from the "Catalogue of the Baldwin Locomotive Works" for 1881, to which have been added some later figures from "Record No. 65" of the same establishment, and also some obtained by the writer directly from the roads concerned. Being taken thus at random, the results may be accepted as fairly representative of American practice.

Attention should be directed to the fact that the performance of the 10-34 E, Consolidation locomotive on the Lehigh Valley Railroad in 1871 is practically equal to that of the latest Mallet compounds on the Great Northern Railway. In other words, in the ratio between the ability to produce steam and the weight on the drivers there has been no change in the last forty years. This would indicate that the figures are not likely to be changed much as long as steam-driven locomotives are in use. What will obtain with the introduction of electric traction is "another story."

These results have also been platted, and are presented in [Fig. 1](#), with the lengths of grade as abscissas and the percentages of weight utilized as ordinates. The curve sketched to represent a general average will show the conditions at a glance. The results may at first sight seem irregular, but the agreement is really remarkable when the variety of sources is considered; that in many cases the "reputed" rate of grade is doubtless given without actual measurement; that the results also include momentum, the ability to utilize which depends on the conditions of grade, alignment, and operating practice which obtain about the foot of each grade; and that the same amount of energy due to momentum will carry a train farther on a light grade than on a heavy one.

There are four items in [Table 1](#) which vary materially from the general consensus. For Item 9, the authorities of the road particularly state that their loads are light, because, owing to the congested condition of their business, their trains must make fast time. Item 10 represents very old practice, certainly prior to 1882, and is "second-hand." The load consisted of empty coal cars, and the line was very tortuous, so that it is quite probable that the resistance assumed in the calculation is far below the actual. Items 15 and 17 are both high. To account for this, it is to be noted that this road has been recently completed, regardless of cost in the matter of both track and rolling stock, and doubtless represents the highest development of railroad practice. Its rolling stock is all new, and is probably in better condition to offer low resistance than it will ever be again, and there were no "foreign" cars in the trains considered. The train resistance, therefore, may be naturally assumed to be much less than that of roads hauling all classes of cars, many of which are barely good enough to pass inspection. As the grades are light in both cases, this feature of train resistance is larger than in items including heavier grades. Attention should be called to the fact that

a line connecting the two points representing these items on [Fig. 1](#) would make only a small angle with the sketched curve, and would be practically parallel to a similar line connecting the points represented by Items 13 and 16. There is, therefore, an agreement of ratios, which is all that needs consideration in this discussion.

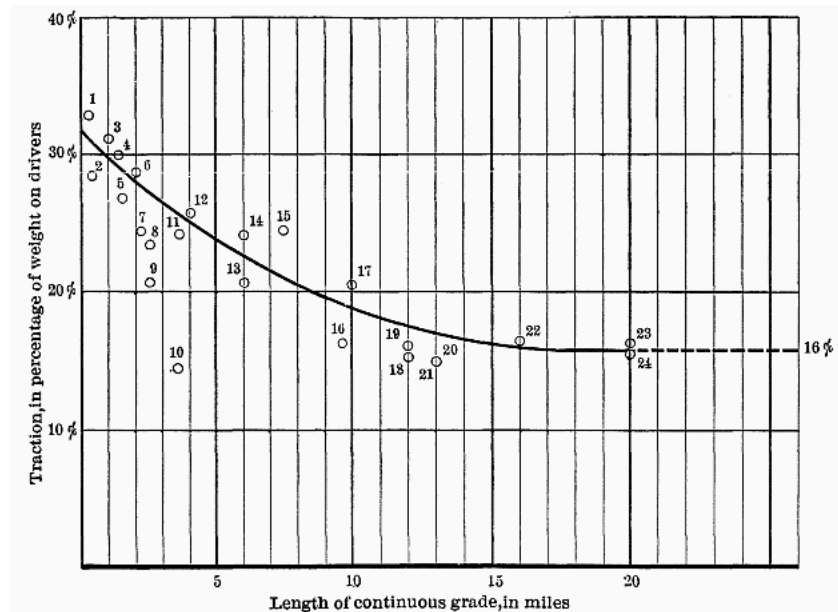


Fig. 1.—DIAGRAM SHOWING PERCENTAGE OF WEIGHT ON DRIVERS WHICH IS UTILIZED IN TRACTION ON GRADES OF VARIOUS LENGTHS

Wellington, in his monumental work on railway location, presents a table of this character. The percentages of weight on the drivers which is utilized in draft show the greatest irregularity. He does not give the length of the grades considered, so that it is impossible to say how far the introduction of this feature would have contributed to bring order out of the chaos. In his discussion of the table he admits the unsatisfactory character of the results, and finally decides on 25% as a rough average, "very approximately the safe operating load in regular service." He further states that a number of results, which he omits for want of space, exceeds 33 per cent. The highest shown in [Table 1](#) will be found in Item 1 (0.06 mile, 0.066 grade), showing 33 per cent. There is no momentum effect here, as the grade is a short incline extending down to the river, and the start is necessarily a "dead" one. The reports of Item 3, which shows 31%, and Item 5, which shows 27%, state specifically that the locomotives will stop and start the loads given at any point on the grade.

The results of a series of experiments reported by Mr. A. C. Dennis in his paper, "Virtual Grades for Freight Trains," previously referred to, indicate a utilization of somewhat more than 23%, decreasing with the speed.

All this indicates that the general failure of locomotives to utilize more than from 16 to 18% on long grades, as shown by [Table 1](#), can only be due to the failure of the boilers to supply the necessary steam. While the higher percentage shown for the shorter grades may be ascribed largely to momentum present when the foot of the grade is reached, the energy due to stored heat is responsible for a large portion of it.

When a locomotive has been standing still, or running with the steam consumption materially below the production, the pressure accumulates until it reaches the point at which the safety valve is "set." This means that the entire machine is heated to a temperature sufficient to maintain this pressure in the boiler. When the steam consumption begins to exceed the production, this temperature is reduced to a point where the consumption and production balance.

The heat represented by this difference in temperature has passed into the steam used, thus adding to the energy supplied by the combustion going on in the furnace. The engines, therefore, are able to do considerably more work during the time the pressure is falling than they can do after the fall has ceased.

The curve in [Fig. 1](#) would indicate that the energy derived from the two sources just discussed is practically dissipated at 15 miles, though the position of the points representing Items 16, 18, 19, 20, and 21 would indicate that this takes place more frequently between 10 and 12 miles. From this point onward the performance depends on the efficiency of the steam production, which does not appear to be able to utilize more than 16% of the weight on the drivers. The diagrams presented by Mr. Dennis in his paper on virtual grades, and by John A. Fulton, M. Am. Soc. C. E., in his discussion of that paper, indicate that similar results would be shown were they extended to include the distance named.

From this it would appear that a locomotive is capable of hauling a larger train on grades less than 10 miles in length than on longer grades, and that, even when unexpectedly stopped, it is capable of starting again as soon as the steam pressure is sufficiently built up. Conversely, it should be practicable to use a higher rate of ascent on shorter grades on any given line without decreasing the load which can be hauled over it. In other words, what is known as the "ruling grade" is a function, strictly speaking, of the length as well as the rate of grade.

In any discussions of the practicability of using a higher rate on the short grades, which the writer has seen, the most valid objection has appeared to be the danger of stalling and consequent delay. As far as momentum is relied on, this objection is valid. Within the limits of the load which can be handled by the steam, it has small value, as it is only a question of waiting a few minutes until the pressure can be built up to the point at which the load can be handled. As this need only be an occasional occurrence, it is not to be balanced against any material saving in cost of construction.

The writer does not know of any experiments which will throw much light on the value of heat storage as separated from momentum, though the following discussion may prove suggestive:

A train moving at a rate of 60 ft. per sec., and reaching the foot of a grade, will have acquired a "velocity head" of 56.7 ft., equivalent to stored energy of $56.7 \times 2,000 = 113,400$ ft.-lb. per ton. On a 0.002 grade (as in Item 15 of [Table 1](#)) the resistance would be, gravity 4 lb. + train 4.7 lb. = 8.7 lb., against which the energy above given would carry the train through $113,400 \div 8.7 = 13,034$ ft., say, 2.5 miles, leaving 5 miles to be provided for by the steam production. Examining the items in the table having grades in excess of 10 miles, it will be noted that 16% is about all the weight on drivers which can be utilized by the current supply of steam. In Item 15 the energy derived from all sources is equivalent to 24.3%; hence the stored heat may be considered as responsible for an equivalent of $24.3\% - 16\% = 8.3\%$ for a distance of 5 miles.

TABLE 1.

Item No.	Length of grade, in miles.	Rate of grade.	Maximum curvature.	Compensation.	Gross weight of load, in tons.	Weight of tender, in tons.	Weight of locomotive, in tons.	Weight on drivers, in tons.	Percentage of weight on drivers utilized in draft.	Class.	Maker.	Railroad.	Reporting Officer.	Year.	Source Data.
1	0.06	0.066			115		37.5	29	0.358	8-28-1/3 C	Baldwin.	Morgan's Louisiana & Texas	Newell Tilton, Asst. Supt.	1880	Baldwin Catalog 1881, p. 134
2	0.33	0.0203	25°20'		242	25	35	23	0.285	8-28 C	"	Long Island	S. Spencer, Gen. Supt.	1878	Baldwin Catalog 1881, p.
3	1.0	0.06	16°	0.05	192	22	57.5	50	0.310	10-36 E	"	Atchison, Topeka & Santa Fe	J. D. Burr, Asst. Engr.	1879	Baldwin Catalog 1881, p. 115
4	1.3	0.0127			600	16	40	32.5	0.300	Mogul.	"	Chillan & Talcahuana	J. E. Martin, Local Supt.	1879	Baldwin Catalog 1881, p. 100
5	1.4	0.0128	3°12'		750	15	51	44	0.270	10-34 E	"	Chicago, Burlington & Quincy	H. B. Stone	1880	Baldwin Catalog 1881, p. 116
6	2.0	0.01			1,000	15	51	44	0.291	10-34 E	"	Chicago, Burlington & Quincy	"	1880	Baldwin Catalog

															1881, p. 116
7	2.2	0.013	3°		725	15	51	44	0.245	10-34 E	"	Chicago, Burlington & Quincy	"	1880	Baldwin Catalog 1881, p. 116
8	2.5	0.0144	6°		400	27	42	32	0.237	10-32 E	"	St. Louis & San Francisco	C. W. Rogers, Gen. Mgr.	1879	Baldwin Catalog 1881, p.
9	2.5	0.004			2,700	70	96.7	85.8	0.207	H 6 - A	Pa. R.R.	Cumberland Valley.		1910	
10	3.5	0.033	14°		100	25	35	35	0.160					1910	Trautwir Pocket Book, Ed. 1882, p. 412
11	3.6	0.035	10°	0.05	236	22	57.5	50	0.245	10-36 E	Baldwin.	Atchison, Topeka & Santa Fe	J. D. Burr, Asst. Engr.	1879	Baldwin Catalog 1881, p. 114
12	4.0	0.0085	4°		1,020	30	51	44	0.256	10-34 E	"	Missouri Pacific	John Hewitt, Supt. M. P.	1880	Baldwin Catalog 1881, p. 112
13	6.0	0.0145			308	25	38	28	0.207	10-28 D	"	Western Maryland	D. Holtz, M. of Mach'y.	1878	Baldwin Catalog 1881, p.
14	6.0	0.020	10°	0.05	460	32	57.5	50	0.242	10-34 E	"	Atchison, Topeka & Santa Fe	J. D. Burr, Asst. Engr.	1879	Baldwin Catalog 1881, p. 114
15	7.5	0.002		C	6,152	86	134.5	109.5	0.243	Mallet.	"	Virginian Ry.		1910	Engineer News, Jan. 13, 1910
16	9.75	0.018			200	18	29	29	0.170			Pennsylvania		1910	Trautwir Pocket Book, Ed. 1882, p. 412
17	10.0	0.006		C	6,173	86	299	265	0.203	Mallet.	Baldwin.	Virginian Ry.		1910	Engineer News, Jan. 13, 1910
18	12.0	0.018	10°		280	30	51	44	0.160	10-34 E	"	Lehigh Valley, Wyoming Div.	A. Mitchell, Div. Supt.	1871	Baldwin Catalog 1881, p. 112
19	12.0	0.022			850	74	175	156	0.166	D-D 16	"	Great Northern	Grafton Greenough.	1908	Baldwin Loco. W Record, No. 65, p. 29
20	13.0	0.022			800	74	177	158	0.153	D-D 1	"	Great Northern	Grafton Greenough.	1908	Baldwin Loco. W Record, No. 65, p. 29
21	13.0	0.022	14°		415	50	91	83	0.154	Consol.	"	Baltimore & Ohio	F. E. Blaser, Div. Supt.	1910	
22	16.0	0.0044			950	30	51	44	0.164	10-34 E	"	Central of N. J.	W. W. Stearns, Asst.Gen.Supt.	1880	Baldwin Catalog 1881, p. 113
23	20.0	0.022			500	62	97.5	90	0.170	F 8, Consol.	"	Great Northern	Grafton Greenough.	1908	Baldwin Loco. W Record, No. 65, p. 29

24	20.0	0.022			800	74	177	158	0.159	L-1, Mallet.	"	Great Northern	Grafton Greenough.	1906	Baldwin Loco. W Record, No. 65, p. 29
----	------	-------	--	--	-----	----	-----	-----	-------	-----------------	---	-------------------	-----------------------	------	---

In proportioning grade resistance for any line, therefore, a locomotive may be counted on to utilize 24.3% of the weight on the drivers for a distance of 5 miles on a 0.002 grade without any assistance from momentum, and, in the event of an unexpected stop, should be able, as soon as a full head of steam is built up, to start the train and carry it over the grade. This is probably a maximum, considering the condition of the equipment of this Virginian Railway, as previously mentioned.

Treating Item 14 in the same way, a distance of 2,310 ft. is accounted for by momentum, leaving, say, 5.5 miles for the steam, or the length of a 0.02 grade on which a locomotive may be loaded on a basis of tractive power equal to 24.2% of the weight on the drivers.

From these figures it may be concluded that on lines having grades from 12 to 15 or more miles in length, grades of 3 to 5 miles in length may be inserted having rates 50% in excess of that of the long grades, without decreasing the capacity of the line. This statement, of course, is general in its bearings, each case being subject to its especial limitations, and subject to detailed calculations.

It may be noted that the velocity of 60 ft. per sec., assumed at the foot of the grade, is probably higher than should be expected in practice; it insures, on the other hand, that quite enough has been allowed for momentum, and that the results are conservative.

Arguments like the foregoing are always more or less treacherous; being based on statistics, they are naturally subject to material modifications in the presence of a larger array of data, therefore, material assistance in reaching practical conclusions can be given by the presentation of additional data.

DISCUSSION

C. D. PURDON, M. AM. SOC. C. E. (by letter).—Some years ago the writer, in making studies for grade revision, found that the tractive power of a locomotive up grade becomes less as the length of the grade increases, and in some unknown proportion. This was a practical confirmation of the saying of locomotive engineers, that the engine "got tired" on long grades. On a well-known Western railroad, with which the writer is familiar, experiments were made for the purpose of rating its locomotives. The locomotives were first divided into classes according to their tractive power, this being calculated by the usual rule, with factors of size of cylinders, boiler pressure, and diameter of drivers, also by taking one-fourth of the weight on the drivers, and using the lesser of the two results as the tractive power.

Locomotives of different classes, and hauling known loads, were run over a freight division, the cars being weighed for the purpose; thus the maximum load which could be handled over a division, or different parts of a division, was ascertained, and this proportion of tonnage to tractive power was used in rating all classes.

Of course, this method was not mathematically accurate, as the condition of track, the weather, and the personal equation of the locomotive engineers all had an effect, but, later, when correcting the rating by tests with dynamometers, it was found that the results were fairly practical.

There were three hills where the rate of grade was the same as the rest of the division, but where the length was much in excess of other grades of the same rate.

Designating these hills as A, B, and C, the lengths are, respectively, 2.44, 3.57, and 4.41 miles. There were no other grades of the same rate exceeding 1 mile.

In one class of freight engines, 10-wheel Brooks, the weight of the engine was 197,900 lb.; tender, 132,800 lb.; weight on drivers, 142,600 lb.; boiler pressure, 200 lb.; and tractive power of cylinders, 33,300 lb.

On Hill A these engines are rated at 865 tons, as compared with 945 on other parts of the division. As the engine weighs 165 tons and the caboose 15 tons, 180 tons should be added, making the figures, 1,045 and 1,125 tons. Thus the length of the grade, 2.44 miles, makes the tractive power on it 92% of that on shorter grades.

On Hill B, the rating, adding 180 tons as above, is 1,160 and 1,230 tons, respectively, giving 94% for 3.57 miles.

On Hill C, the rating, with 180 tons added, is 1,130 and 1,230 tons, making 92% for 4.41 miles.

Taking the same basis as the author, namely, 4.7 lb. per ton, rate of grade $\times$ 20, and weight on drivers, gives:

Hill A, 18.078%, remainder of division, 19.462%
Hill B, 20.068%, remainder of division, 21.279%
Hill C, 19.549%, remainder of division, 21.279%

It will be noted that the author uses the weight on the drivers as the criterion, but the tractive power is not directly as the weight on the drivers, some engines being over-cylindereed, or under-cylindereed; in the class of engines above mentioned the tractive power is 23.35% of the weight on the drivers.

The writer made a study of several dynamometer tests on Hill C. There is a grade of the same rate, about 1 mile long, near this hill, and a station near its foot, but there is sufficient level grade between this station and the foot of the hill to get a good start.

All the engines of the above class, loaded for Hill C, gained speed on the 1-mile grade, but began to fall below the theoretical speed at a point about 2-1/4 miles from the foot of the hill. This condition occurred when the trains stopped at the station and also when they passed it at a rate of some 16 or 18 miles per hour, the speed becoming less and less as the top of the hill was approached.

The writer concludes that the author might stretch his opinion as to using heavier rates of grade on shorter hills than 10 miles, and indeed his diagram seems to intimate as much, and that, for economical operation, the

maximum rate of grade should be reduced after a length of about 2 miles has been reached, and more and more in proportion to the length of the hill, in order that the same rating could be applied all over a division.

This conclusion might be modified by local conditions, such as an important town where cars might be added to or taken from the train.

While it does not seem practicable to the writer to calculate what the reduction of rate of grade should be, a consensus of results of operation on different lengths of grade might give sufficient data to reach some conclusion on the matter.

The American Railway Engineering and Maintenance of Way Association has a Committee on "Railway Economics," which is studying such matters, but so far as the writer knows it has not given this question any consideration.

The writer hopes that the author will follow up this subject, and that other members will join, as a full discussion will no doubt bring some results on a question which seems to be highly important.

JOHN C. TRAUTWINE, JR., ASSOC. AM. SOC. C. E. (by letter).—In his collection of data, Mr. Randolph includes two ancient cases taken from the earliest editions (1872-1883) of Trautwine's "Civil Engineer's Pocket-Book," referring to performances on the Mahanoy and Broad Mountain Railroad (now the Frackville Branch of the Reading) and on the Pennsylvania Railroad, respectively.

In the private notes of John C. Trautwine, Sr., these two cases are recorded as follows:

"On the Mahanoy & Broad Mtn. R. R., *tank* Engines of 35 tons, *all on 8 drivers*, draw 40 *empty* coal cars weighing 100 tons, *up* a continuous grade of 175 ft. per mile for 3-1/2 miles; & around curves of 450, 500, 600 ft. &c. rad., at 8 miles an hour. (1864) This is equal to 77-14/100 tons for a 27-ton engine." (Vol. III, p. 176.)

"On the Penn Central 95 ft. grades for 9-3/4 miles, a 29-ton engine all on 8 drivers takes 125 tons of freight and 112 tons of engine, tender, & cars, in all 237 tons,^[C] and a passenger engine takes up 3 cars at 24 miles an hour (large 8 wheels). When more than 3, an auxiliary engine."

It will be seen that Mr. Randolph is well within bounds in ascribing to the Mahanoy and Broad Mountain case (his No. 10) a date "certainly prior to 1882," the date being given, in the notes, as 1864; while another entry just below it, for the Pennsylvania Railroad case, is dated 1860.

It also seems, as stated by Mr. Randolph, quite probable that the frictional resistance (6 lb. per 2,000 lb.) assumed by him in the calculation is far below the actual for this Case 10. The small, empty, four-wheel cars weighed only 4,400 lb. each. Furthermore, the "tons," in the Trautwine reports of these experiments, were tons of 2,240 lb. On the other hand, the maximum curvature was 12° 45' (not 14°, as given by the author), and the engine was a tank locomotive, whereas the author has credited it with a 25-ton tender.

After making all corrections, it will be found that, in order to bring the point, for this Case 10, up to the author's curve, instead of his 6 lb. per 2,000 lb., a frictional resistance of 66 lb. per 2,000 lb. would be required, a resistance just equal to the gravity resistance on the 3.3% grade, making a total resistance of 132 lb. per 2,000 lb.

While this 66 lb. per ton is very high, it is perhaps not too high for the known conditions, as above described. For modern rolling stock, Mr. A. K. Shurtleff gives the formula:^[D]

$$\left. \begin{array}{l} \text{Frictional resistance, on tangent,} \\ \text{in pounds per 2,000 pounds} \end{array} \right\} = 1 + 90 \div C,$$

where C = weight of car and load, in tons of 2,000 lb. This would give, for 4,400-lb. (2.2-ton) cars, a frictional resistance of 42 lb. per 2,000 lb.; and, on the usual assumption of 0.8 lb. per 2,000 lb. for each degree of curvature, the 12.75° curves of this line would give 10 lb. per ton additional, making a total of 52 lb. per 2,000 lb. over and above grade resistance, under modern conditions.

In the 9th to 17th editions of Trautwine (1885-1900), these early accounts were superseded by numerous later instances, including some of those quoted by the author.

In the 18th and 19th editions (1902-1909) are given data respecting performances on the Catawissa Branch of the Reading (Shamokin Division) in 1898-1901. These give the maximum and minimum loads hauled up a nearly continuous grade of 31.47 ft. per mile (0.59%) from Catawissa to Lofty (34.03 miles) by engines of different classes, with different helpers and without helpers.

[Table 2](#) (in which the writer follows the author in assuming frictional resistance at 4.7 lb. per 2,000 lb.) shows the cases giving the maximum and minimum values of the quantity represented by the ordinates in the author's diagram, namely, "Traction, in percentage of weight on drivers."

It will be seen that the maximum percentage (16.1) is practically identical with that found by the author (16) for grade lengths exceeding 17 miles.

Near the middle of the 34-mile distance there is a stretch of 1.51 miles, on which the average grade is only 5.93 ft. per mile (0.112%), and this stretch divides the remaining distance into two practically continuous grades, 19.39 and 13.13 miles long, respectively; but, as the same loads are hauled over these two portions by the same engines, the results are virtually identical, the maxima furnishing two more points closely coinciding with the author's diagram.

TABLE 2.—TRACTION FORCE, CATAWISSA TO LOFTY.

Length of grade, in miles					34.03
Grade				in feet per mile	31.47
				percentage	A 0.597
Resistances, in pounds per 2,000 lb., Gravity (=20 A) = 11.94. Friction = 4.70				B	16.64
Load:	Cars.	Locomotive.	Tender.		
Maximum ^[E]	1,561	44.60	25.25	C	1,631
Minimum ^[E]	1,031	60.50	34.50	C	1,126
Traction (= B C ÷ 2,000)		Maximum ^[E]		D	13.60
		Minimum ^[E]		D	9.38
Weight on Drivers:		Locomotive.	Helper.		
Maximum ^[E]		21.60	63.00	E	84.60
Minimum ^[E]		47.00	72.00	E	119.00
Percentage (= D ÷ E).					
Maximum				F	16.1
Minimum				F	7.9

^[E] Giving maximum values of percentage, F.

^[E] Giving minimum values of percentage, F.

BEVERLY S. RANDOLPH, M. AM. SOC. C. E. (by letter).—The percentages given by Mr. Purdon would seem to indicate that the length of the grades did not affect the loads in the cases cited, but these percentages are so much below those shown in the table, for similar distances, as to indicate some special conditions which the writer has been unable to find in the text.

The use of the percentage of weight on drivers which is utilized in traction as a measure of the efficiency of the locomotive, while, probably, not applicable to individual machines, is sound for the purposes of comparison of results to be obtained on various portions of a line as far as affected by conditions of grade and alignment. It has the advantage of disregarding questions of temperature, condition of track, character of fuel, etc., which, being the same on all portions of the line, naturally balance and do not affect the comparison. It is, of course, simply a method of expressing the final efficiency of the various parts of the locomotive, and, since it depends entirely on actual results already accomplished, leaves no room for difference of opinion or theoretical error.

The writer has always considered an "under-cylindere" locomotive as a defective machine. All weight is a distinct debit, in the shape of wear and tear of track and running gear, resistance due to gravity on grades, interest on cost, etc. When this weight fails to earn a credit in the way of tractive efficiency, it should not be present.

The statement relative to the performance of locomotives on "Hill C" is interesting, especially in that it appears to have been immaterial whether they made a dead start after stopping at the station or approached the foot of the hill at 16 to 18 miles per hour. The momentum would appear to be an insignificant factor.

It is gratifying to note that Mr. Trautwine has been able to brace up the weak member of [Table 1](#) so completely with his detailed data; also that his other results strengthen the conclusions reached in the paper.

FOOTNOTES:

[A] "The Economic Theory of Railway Location," 1887 edition, p. 502.

[B] *Transactions*, Am. Soc. C. E., Vol. L, p. 1.

[C] "Nearly 200 tons *exclusive* of eng. & ten." (Vol. III, p. 176-1/10.)

[D] American Railway Engineering and Maintenance of Way Association, Bulletin 84, February, 1907, p. 99.

*** END OF THE PROJECT GUTENBERG EBOOK TRANSACTIONS
OF THE AMERICAN SOCIETY OF CIVIL ENGINEERS, VOL. LXX,
DEC. 1910 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project GutenbergTM License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.